Virtual Cloud Gong—Virtual musical instrument performance based on VR glasses

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I. Starting Point

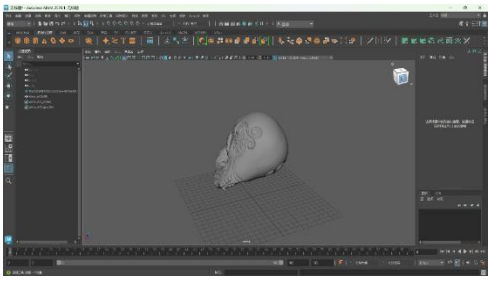
This program was created for the Virtual Musical Instrument Museum project of the MuseBit team I'm in. The Virtual Musical Instrument Museum is an educational and innovative technology project. It is committed to presenting unique musical instrument collections from around the world through information technology. In the virtual space created by the game engine, users can get close to various musical instruments, listen to time - honored melodies, and explore music cultures without leaving their homes.

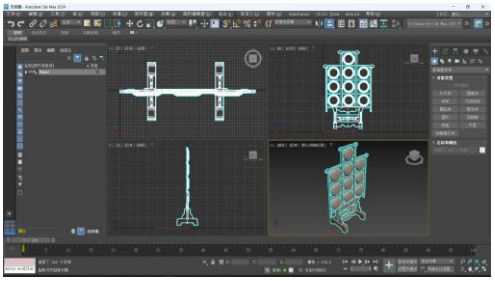
On this basis, I hope that through the virtual performance of musical instruments based on VR glasses, learners can have a clearer, more intuitive, and vivid experience of musical instruments.

II. Production Report

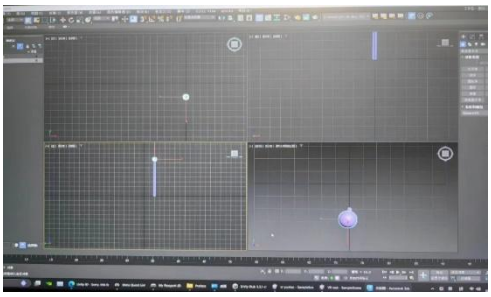
1) Model Inspection

Before starting the program production, I first selected several percussion instruments with high feasibility for virtual performance from the instrument model library of the Virtual Musical Instrument Museum, such as wooden fish, cymbals, and bells. I used the software Maya to check whether their models were correct and whether their components were independent to ensure further production. During the inspection, I found modeling errors in the wooden fish and that the models of different sound - producing bodies of some instruments were not independent. After ruling them out one by one, I decided to choose the ten - tone cloud gong model, which had relatively fewer problems.





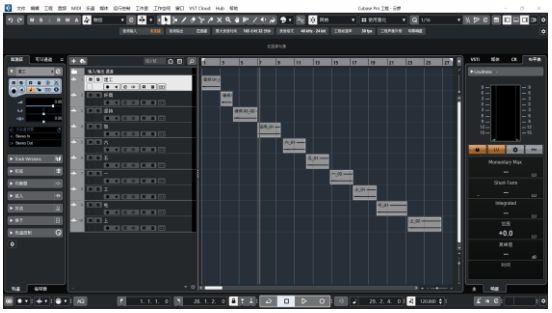
This model only lacked the mallet. After researching information about the mallet, I modeled one independently using 3ds Max. Subsequently, I used the software Blender to modify the model format to fbx files uniformly, which was convenient for later import into Unity.



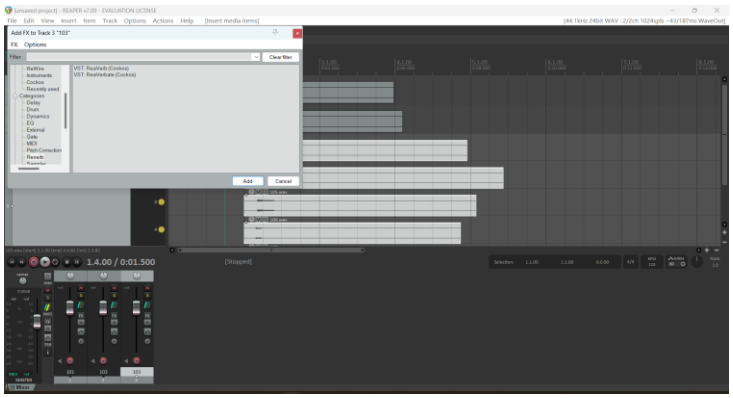
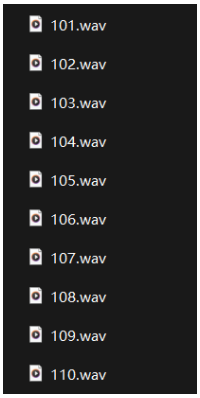
2) Recording and Audio Editing

In the first ten - day period of May, I contacted Senior Guo Xiaoyu from the Modern Percussion Department and went to the basement of Fenyang Road to record the sound sources. Since the ten - tone cloud gong and its relevant materials are rare and different from the cloud gong provided by the school, we consulted a large amount of materials on - site to determine the ten pitches to be recorded. The following are the software and hardware used for recording:

Daw	Sound Card	Microphone
Cubase12	YAMAHA Ag03	Shure sm58



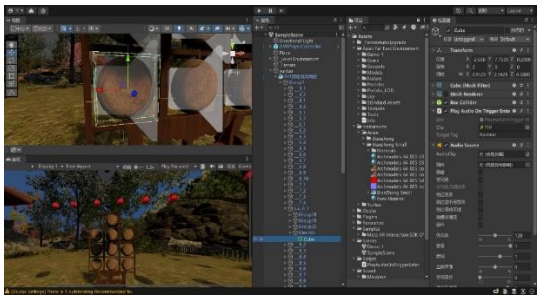
After the recording was completed, I used Reaper for audio editing and tuning. I adjusted the volume to be consistent and added effects to individual audio files with recording flaws.

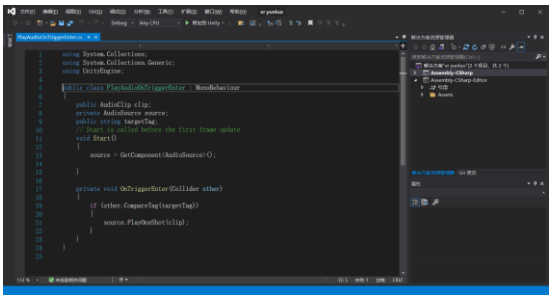



3) Program Coding

The 3D scene of the program was sourced from the AssetStore of Unity's official material store. After preparing all the materials, I imported them into Unity. I added a cube collision box to each gong model, and added scripts for "collision - triggered sound effects", components such as box collider and audio source to the collision box, and attached the

corresponding sound sources.





The device I used was the Oculus Meta Quest 3. After completing the instrument program, I used the handle adaptation script provided by the headset manufacturer to code the program for the right - hand mallet, adjusting its position so that it could be held in the right hand from the player's perspective. I also coded a script for the left - hand to control the character's movement. After importing the program into the headset, I entered the final testing and refinement stage. During this stage, I adjusted details such as the scale of the cloud gong model and the lighting in the scene.

III. Creative Insights

The Virtual Musical Instrument Museum project fully embodies the interdisciplinary nature, which can effectively promote the professional construction of comprehensive universities and art colleges, and facilitate interdisciplinary communication. Under this background, the production of virtual musical instruments further clarifies the project's action guidelines of focusing on reality, assisting teaching, and enhancing the sense of experience.

There is an important reason why I chose to make the virtual ten - tone cloud gong. This kind of percussion instrument is special. Each strike will cause more or less deformation, which will wear the instrument and lead to pitch changes, and this cannot be avoided or controlled. Replicating it in the virtual world can perfectly solve this problem.

In my future studies, I plan to improve the instrument model to make it easier to learn, and add functions such as pitch identification and detailed instrument information. I sincerely hope that the production of the virtual ten - tone cloud gong can inspire the digital action in music education and explore new forms of inheriting Chinese traditional musical instruments.

June 28, 2024